

COMPUTER SCIENCE MSC CURRICULUM

Program: Cybersecurity Specialization

ASSESSMENT TYPES:

E: Exam | PG: Practice Grade | XE: Lecture+Practice with Exam | XPG: Lecture+Practice with Practical Grade

=== CORE COURSES (13 Credits) ===

--- SEMESTER 1 (5 Credits) ---

IPM-22IRMEG: Research methodology

1 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 5 Credits

--- SEMESTER 2 (8 Credits) ---

IPM-22IASTE: Advanced Software Technology (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits

IPM-22IDAAE: Design and analysis of algorithms (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits

IPM-22IPRG: Internship

0 Credits | Semesters: 2-4 | 240 hours

=== COMPULSORY SPECIALIZATION COURSES (29 Credits) ===

--- SEMESTER 1 (17 Credits) ---

IPM-22BACCE: Cryptography (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 2 Credits | Prerequisite: IPM-22BACCG (weak)

IPM-22BACCG: Cryptography (Practice)

0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22BMCSE: Introduction to Computer Security (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 0 Consultation | E | 3 Credits | Prerequisite: IPM-22BMCSG (weak)

IPM-22BMCSG: Introduction to Computer Security (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

IPM-22BADSEG: Introduction to Data Science

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

--- SEMESTER 2 (6 Credits) ---

IPM-22BASQTE: Software quality and testing (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22BASQTG (weak)

IPM-22BASQTG: Software quality and testing (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

--- SEMESTER 3 (6 Credits) ---

IPM-22RACERPE: Cryptographic protocols (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22RACERPG (weak)

IPM-22RACERPG: Cryptographic protocols (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits

=== COMPULSORY ELECTIVE COURSES (Select 42 Credits) ===

--- SEMESTER 1 ELECTIVES (8 Credits) ---

IPM-22RAPCMMSG: Preparation course for master studies and developing learning skills

0 Lecture, 3 Practice, 0 Laboratory, 0 Consultation | PG | 2 Credits

IPM-22RADFISE: Development of Financial IT Systems (Lecture)

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Semesters: 1,3 | Prerequisite: IPM-22RADFISG (weak)

IPM-22RADFISG: Development of Financial IT Systems (Practice)

0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits | Semesters: 1,3

--- SEMESTER 2 ELECTIVES (16 Credits) ---

IPM-22RASPEG: Information security and privacy

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-22RAMLEG: Machine Learning

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits | Prerequisite: IPM-22BADSEG
IPM-22RANSSE: Network and System Security (Lecture)
2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22RANSSG (weak)
IPM-22RANSSCG: Network and System Security (Practice)
0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits
IPM-22RADDSEG: Numerical Methods for Optimization
2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XPG | 6 Credits
IPM-22RAPQCE: Post-quantum cryptography (Lecture)
2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits
IPM-22RATCG: Topics in cryptography seminar
0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits

--- SEMESTER 3 ELECTIVES (18 Credits) ---

IPM-22RACRE: Advanced cryptography (Lecture)
2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22RACRG (weak)
IPM-22RACRG: Advanced cryptography (Practice)
0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits
IPM-22RSCLAB1: Cyber Security Lab I
0 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | PG | 4 Credits
IPM-22RSCLAB2: Cyber Security Lab II
0 Lecture, 0 Practice, 4 Laboratory, 2 Consultation | PG | 6 Credits
IPM-22RACRAE: Provably secure modular design of cryptographic protocols (Lecture)
2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits | Prerequisite: IPM-22RACRAG (weak)
IPM-22RACRAG: Provably secure modular design of cryptographic protocols (Practice)
0 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | PG | 3 Credits

=== INNOVATION & ENTREPRENEURSHIP MODULE (Optional) ===

IPM-22IREIBEG: I&E Basics
2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits | Semester: 1
IPM-22IREBDL1G: Business Development Lab I
0 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | PG | 4 Credits | Semester: 1
IPM-22IREBDL2G: Business Development Lab II
0 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | PG | 4 Credits | Semester: 2
IPM-22IREIAOEEG: Innosocial aspects of entrepreneurship
2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits | Semester: 2
IPM-22IRETSSG: Thematic Summer Schools with I&E project
1 Lecture, 1 Practice, 0 Laboratory, 2 Consultation | XPG | 4 Credits | Semester: 2
IPM-22IRESTEG: I&E Study
2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits | Semester: 3

=== THESIS & ADDITIONAL ===

IPM-20THCONS: Thesis consultation
30 Credits | Semester: 4

IPM-22ERRASMUS: Erasmus mobility
Maximum 24 Credits | Semester: 3

Optional Course: 6 Credits | Semester: 3

=== CREDIT SUMMARY ===

Core Courses: 13 Credits
Compulsory Specialization Courses: 29 Credits
Compulsory Elective Courses: 42 Credits
Optional Course: 6 Credits
Thesis: 30 Credits
TOTAL DEGREE CREDITS: 120 Credits

Semester Credit Distribution:
Semester 1: 30 | Semester 2: 30 | Semester 3: 30 | Semester 4: 30

SPECIAL NOTES:

- EIT students must complete I&E module and all subjects marked with (*)
- Cybersecurity students fulfill internship requirement by completing Cyber Security Lab I and II
- EIT students complete internship and thesis work in their final semester